

The image of R'_n on the standard hook $V_{(n-1,1)}$, and a clean recurrence for $\text{rank } \Pi^{S_n}|_{V_{(n-1,1)}}$

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8 May 2026 (evening)

Abstract

Companion to 2026-05-08-hook-rank-formula.tex, which established the same lemmas via the D_k dimension sequence. This note re-derives the same content via an alternative recurrence abstraction $\rho_n = 1 + \sigma_{n-1}$.

The staircase Bott–Samelson element Π^{S_n} has the recursive form $\Pi^{S_n} = \Pi^{S_{n-1}} R'_n$, where $R'_n = (T_{n-1}+1)(T_{n-2}+1) \cdots (T_1+1)$. We give a fully structural description of the image of R'_n on the hook cell module $V_{(n-1,1)}$:

$$R'_n(V_{(n-1,1)}) = \text{span}(v_{T^{(2)}}, \dots, v_{T^{(n-2)}}, \alpha_{n-1} v_{T^{(n-1)}} + \beta_{n-1} v_{T^{(n)}}),$$

an $(n-2)$ -dimensional subspace where $T^{(k)}$ is the SYT of $(n-1,1)$ with k in row 2 (entries arranged in increasing order) and $(\alpha_{n-1}, \beta_{n-1})$ are the Hoefsmit constants for the T_{n-1} -block at axial distance $n-1$. As a consequence we obtain the clean rank recurrence

$$\rho_n := \text{rank } \Pi^{S_n}|_{V_{(n-1,1)}} = 1 + \sigma_{n-1}, \quad \sigma_m := \text{rank } \Pi^{S_m}|_{W^{(m)}},$$

where $W^{(m)} \subset V_{(m-1,1)}$ at S_m is the codim-1 subspace $\text{span}(v_{T^{(k)}} : k = 2, \dots, m-1)$ (omitting $v_{T^{(m)}}$). The recurrence is verified at $n \leq 7$, where it yields the conjectural rank formula

$$\rho_n = \left\lceil \frac{n-2}{2} \right\rceil \quad (\text{verified for } n \leq 7).$$

A general structural proof of the rank formula remains open. We give it the alternative formulation: the formula is equivalent to the parity-dependent kernel-containment statement $K_{n-1} \subseteq W^{(n-1)} \iff n \text{ even}$, where $K_{n-1} = \ker \Pi^{S_{n-1}}|_{V_{(n-2,1)}}$. This matches the $\delta(n)$ dichotomy of the morning’s companion writeup but re-cast as a clean dimension-of-intersection statement.

1 Setup and notation

The Hecke algebra and seminormal basis. Let $H_q(S_n)$ be the Iwahori–Hecke algebra of type A_{n-1} with generators $T_1, \dots, T_{n-1}$ subject to $T_i^2 = (q-1)T_i + q$ and the braid relations. For each partition $\lambda \vdash n$, let V_λ be the irreducible cell module (Specht module). We work in the seminormal basis $\{v_T\}$ indexed by standard Young tableaux T of shape λ . For each generator T_i and each SYT T :

- $T_i v_T = q v_T$ if $i, i+1$ lie in the same row of T (axial distance $d = 1$).
- $T_i v_T = -v_T$ if $i, i+1$ lie in the same column ($d = -1$).

- Otherwise ($|d| \geq 2$), T_i acts as a 2×2 block on $\text{span}(v_T, v_{s_i T})$ with eigenvalues q and -1 . The $+q$ -eigenvector in this block is $\alpha_d v_T + \beta_d v_{s_i T}$ for nonzero Hoefsmit constants α_d, β_d (which depend on d but not on T).

The element $(T_i + 1)$ acts on V_λ as: $(1+q) \cdot \text{id}$ on the $+q$ -eigenspace; 0 on the -1 -eigenspace; and on a 2×2 block, as the rank-1 operator that projects onto the $+q$ -eigenvector and scales by $(1+q)$.

The staircase Bott–Samelson and R'_k . The descending staircase reduced word for $w_0 \in S_n$ is $[1, 2, 1, 3, 2, 1, \dots, n-1, n-2, \dots, 1]$. The Bott–Samelson element associated to this word is

$$\Pi^{S_n} = \prod_{k=1}^{n-1} (T_{k+1}+1)(T_{k-1}+1) \cdots (T_1+1) = R'_2 R'_3 \cdots R'_n,$$

with the convention that products act on vectors right-to-left. Here $R'_k := (T_{k-1}+1)(T_{k-2}+1) \cdots (T_1+1)$ for $k \geq 2$. In particular $\Pi^{S_n} = \Pi^{S_{n-1}} R'_n$, since $\Pi^{S_{n-1}} = R'_2 \cdots R'_{n-1}$ uses only generators $T_1, \dots, T_{n-2}$.

The hook $(n-1, 1)$ and its seminormal basis. For the partition $\lambda = (n-1, 1)$ at S_n , the SYTs are indexed $T^{(k)}$ for $k = 2, 3, \dots, n$, where $T^{(k)}$ is the unique SYT with k in row 2 (at position $(2, 1)$) and the remaining entries $\{1, \dots, n\} \setminus \{k\}$ filling row 1 in increasing order. We write $v_k := v_{T^{(k)}}$ throughout. Thus $\dim V_{(n-1, 1)} = n - 1$.

Action of T_j on $V_{(n-1, 1)}$. For $T^{(k)}$, the position of entry i has content $i - 1$ if $i < k$; content -1 if $i = k$; content $i - 2$ if $i > k$. Hence:

- T_1 has eigenvalue -1 on v_2 (column case) and $+q$ on v_k for $k \geq 3$ (row case).
- For $j \geq 2$, T_j has a 2×2 block on $\text{span}(v_j, v_{j+1})$ at axial distance $\pm j$, and acts as $+q$ on v_k for $k \notin \{j, j+1\}$.

We write α_j, β_j for the Hoefsmit constants in the $+q$ -eigenvector $\alpha_j v_{T^{(j)}} + \beta_j v_{T^{(j+1)}}$ at axial distance j (where $v_{T^{(j)}} = v_j$, etc.). *Both α_j and β_j are nonzero:* otherwise $T_j v_j$ would be a scalar multiple of v_j , contradicting the 2×2 block structure.

2 The image of R'_n on $V_{(n-1, 1)}$

2.1 Statement

Theorem 1 (R'_n -image lemma). *For $n \geq 3$,*

$$R'_n(V_{(n-1, 1)}) = \text{span}(v_2, v_3, \dots, v_{n-2}, \alpha_{n-1} v_{n-1} + \beta_{n-1} v_n), \quad (1)$$

an $(n-2)$ -dimensional subspace of $V_{(n-1, 1)}$. Equivalently, R'_n is rank- $(n-2)$ on $V_{(n-1, 1)}$ with kernel $\ker R'_n = \text{span}(v_2)$.

2.2 Proof: induction over the partial product

We prove a stronger statement by induction on the length of the partial product. Define

$$P_j := (T_j+1)(T_{j-1}+1) \cdots (T_1+1), \quad 1 \leq j \leq n-1.$$

So $R'_n = P_{n-1}$.

Lemma 2 (Inductive image formula). *For $1 \leq j \leq n-1$,*

$$P_j(V_{(n-1,1)}) = \text{span}(v_2, v_3, \dots, v_{j-1}, \alpha_j v_j + \beta_j v_{j+1}, v_{j+2}, \dots, v_n), \quad (2)$$

an $(n-2)$ -dimensional subspace.

Proof. We proceed by induction on j .

Base case ($j = 1$). $P_1 = T_1 + 1$. Since $T_1 v_2 = -v_2$ and $T_1 v_k = qv_k$ for $k \geq 3$: $P_1 v_2 = 0$, and $P_1 v_k = (1+q)v_k$ for $k \geq 3$. Hence $P_1(V_{(n-1,1)}) = \text{span}(v_3, v_4, \dots, v_n)$.

The right-hand side of (2) at $j = 1$ is $\text{span}(\alpha_1 v_1 + \beta_1 v_2, v_3, \dots, v_n)$. There is no v_1 , so we interpret $j = 1$ via the convention that the leading combination $\alpha_1 v_1 + \beta_1 v_2$ is replaced by “ v_3 in, v_2 out” — equivalently, the formula reads $\text{span}(v_3, v_4, \dots, v_n)$. *This base case is the “boundary” instance, where the formula is cleaner if we describe $P_1(V) = +q\text{-eig}(T_1)$ directly.*

Inductive step. Assume the formula for P_{j-1} , and apply $(T_j + 1)$ to each generator.

For each generator g of $P_{j-1}(V)$ from (2) at level $j-1$:

- $g = v_k$, $2 \leq k \leq j-2$: in $T^{(k)}$, both j and $j+1$ lie in row 1 (since $k < j$, so k is in row 2 and $j, j+1$ are in row 1). Therefore $T_j v_k = qv_k$ and $(T_j+1)v_k = (1+q)v_k$.
- $g = \alpha_{j-1}v_{j-1} + \beta_{j-1}v_j$: Decompose by linearity.
 - $T_j v_{j-1}$: in $T^{(j-1)}$, $j-1$ is in row 2, both j and $j+1$ are in row 1 (positions $(1, j-1)$ and $(1, j)$ respectively). Same row $\Rightarrow T_j v_{j-1} = qv_{j-1}$, and $(T_j+1)v_{j-1} = (1+q)v_{j-1}$.
 - $T_j v_j$: in $T^{(j)}$, j is in row 2 at $(2, 1)$ (content -1) and $j+1$ is in row 1 at $(1, j)$ (content $j-1$). Axial distance $d = j$. So T_j acts on $\text{span}(v_j, v_{j+1})$ as a 2×2 block. By definition of the Hoefsmit constants, $(T_j+1)v_j = c_j(\alpha_j v_j + \beta_j v_{j+1})$ for a nonzero scalar $c_j \in \mathbb{Q}(q)^\times$.

Hence $(T_j+1)g = \alpha_{j-1}(1+q)v_{j-1} + \beta_{j-1}c_j(\alpha_j v_j + \beta_j v_{j+1})$.

- $g = v_{j+1}$: in $T^{(j+1)}$, $j+1$ is in row 2 at $(2, 1)$, j is in row 1 at $(1, j)$. Axial distance $d = -j$. So $(T_j+1)v_{j+1} = c'_j(\alpha_j v_j + \beta_j v_{j+1})$ for a nonzero scalar $c'_j \in \mathbb{Q}(q)^\times$.
- $g = v_k$, $k \geq j+2$: j and $j+1$ both lie in row 1 of $T^{(k)}$ (positions $(1, j)$ and $(1, j+1)$). Same row $\Rightarrow (T_j+1)v_k = (1+q)v_k$.

Collecting: $(T_j+1)P_{j-1}(V)$ is spanned by

$$\{(1+q)v_2, \dots, (1+q)v_{j-2}, \alpha_{j-1}(1+q)v_{j-1} + \beta_{j-1}c_j w_j, c'_j w_j, (1+q)v_{j+2}, \dots, (1+q)v_n\},$$

where $w_j = \alpha_j v_j + \beta_j v_{j+1}$.

Subtracting $\frac{\beta_{j-1}c_j}{c'_j} \cdot c'_j w_j$ from the “ $g = \alpha_{j-1}v_{j-1} + \beta_{j-1}v_j$ ”-image yields the cleaner generator $\alpha_{j-1}(1+q)v_{j-1}$ (proportional to v_{j-1}). After normalizing each generator by removing nonzero scalar multiples, we obtain

$$P_j(V) = \text{span}(v_2, v_3, \dots, v_{j-1}, \alpha_j v_j + \beta_j v_{j+1}, v_{j+2}, \dots, v_n),$$

which is exactly (2). The dimension is $(j-2) + 1 + (n-j-1) = n-2$, independent of j . □

Proof of Theorem 1. $R'_n = P_{n-1}$, so the claim is the $j = n-1$ case of Lemma 2, which yields exactly (1). The kernel computation is immediate: $\dim V_{(n-1,1)} - \dim \text{Im}(R'_n) = (n-1) - (n-2) = 1$, and $v_2 \in \ker R'_n$ since $(T_1+1)v_2 = 0$ (the rightmost factor kills v_2). □

2.3 Computational verification

The image computation in Lemma 2 is verified at $n \in \{3, 4, 5, 6, 7\}$ using the Hoefsmit-basis script `verify_Rn_image.py`. For each n : $\dim R'_n(V_{(n-1,1)}) = n - 2$; the unit vectors $v_2, v_3, \dots, v_{n-2}$ all lie in the image; and the projection of the image onto the $\text{span}(v_{n-1}, v_n)$ coordinates has rank exactly 1 (confirming the single mixing line $\alpha_{n-1}v_{n-1} + \beta_{n-1}v_n$).

3 The rank recurrence

3.1 Statement

Define

$$\rho_n := \text{rank } \Pi^{S_n}|_{V_{(n-1,1)}}, \quad \sigma_m := \text{rank } \Pi^{S_m}|_{W^{(m)}},$$

where $W^{(m)} := \text{span}(v_{T(k)} : 2 \leq k \leq m-1) \subset V_{(m-1,1)}$ at S_m is the codim-1 subspace omitting $v_{T(m)}$. We have $\dim W^{(m)} = m - 2$.

Theorem 3 (Rank recurrence). *For $n \geq 3$,*

$$\rho_n = 1 + \sigma_{n-1}. \quad (3)$$

Proof. By $\Pi^{S_n} = \Pi^{S_{n-1}} R'_n$ and Theorem 1,

$$\text{Im}(\Pi^{S_n}|_{V_{(n-1,1)}}) = \Pi^{S_{n-1}}(R'_n(V_{(n-1,1)})) = \Pi^{S_{n-1}}(Y_n),$$

where $Y_n = \text{span}(v_2, \dots, v_{n-2}, \alpha_{n-1}v_{n-1} + \beta_{n-1}v_n)$.

Branching at S_{n-1} . The branching $V_{(n-1,1)} \downarrow S_{n-1} = V_{(n-2,1)} \oplus V_{(n-1)}$ is multiplicity-free. In the seminormal basis, $V_{(n-2,1)} = \text{span}(v_k : 2 \leq k \leq n-1)$ (the SYTs where n sits in row 1), and $V_{(n-1)} = \text{span}(v_n)$ (the SYT where n sits in row 2). On these summands $\Pi^{S_{n-1}}$ acts as follows:

- On $V_{(n-2,1)}$: as $\Pi^{S_{n-1}}|_{V_{(n-2,1)}}$, the standard hook-rep operator at S_{n-1} .
- On $V_{(n-1)}$ (trivial rep): as the scalar $(1+q)^{\binom{n-1}{2}}$, since each T_j acts as q on the trivial rep, so each (T_j+1) acts as $(1+q)$.

Decomposing Y_n along the branching. Using the seminormal basis: $v_k \in V_{(n-2,1)}$ for $k \leq n-1$, and $v_n \in V_{(n-1)}$. Hence

$$Y_n = W^{(n-1)} \oplus \text{span}(\alpha_{n-1}v_{n-1} + \beta_{n-1}v_n),$$

where $W^{(n-1)} = \text{span}(v_2, \dots, v_{n-2}) \subset V_{(n-2,1)}$ is the codim-1 subspace from the recurrence's definition. The summand $W^{(n-1)}$ lies entirely in $V_{(n-2,1)}$; the second summand has both a $V_{(n-2,1)}$ -component (the $\alpha_{n-1}v_{n-1}$ term) and a $V_{(n-1)}$ -component (the $\beta_{n-1}v_n$ term).

Image dimension. Apply $\Pi^{S_{n-1}}$ to each summand of Y_n :

$$\begin{aligned} \Pi^{S_{n-1}}(W^{(n-1)}) &\subseteq V_{(n-2,1)}, \\ \Pi^{S_{n-1}}(\alpha_{n-1}v_{n-1} + \beta_{n-1}v_n) &= \alpha_{n-1} \Pi^{S_{n-1}}(v_{n-1}) + \beta_{n-1} (1+q)^{\binom{n-1}{2}} v_n. \end{aligned}$$

The first lies in $V_{(n-2,1)}$; the second is a sum of a $V_{(n-2,1)}$ -vector and a nonzero scalar multiple of v_n (since $\beta_{n-1} \neq 0$ and the trivial-rep scalar is nonzero).

Set $A := \Pi^{S_{n-1}}(W^{(n-1)}) \subset V_{(n-2,1)}$ and $y := \alpha_{n-1} \Pi^{S_{n-1}}(v_{n-1}) + \beta_{n-1} (1+q)^{\binom{n-1}{2}} v_n$. Then $\Pi^{S_{n-1}}(Y_n) = A + \text{span}(y)$.

The crucial observation: y 's $V_{(n-1)}$ -component is nonzero ($= \beta_{n-1} (1+q)^{\binom{n-1}{2}} v_n \neq 0$), while $A \subset V_{(n-2,1)}$ has $V_{(n-1)}$ -component zero. Hence $y \notin A$, and so

$$\dim(A + \text{span}(y)) = \dim A + 1.$$

Therefore $\rho_n = \dim \Pi^{S_{n-1}}(Y_n) = \dim A + 1 = \sigma_{n-1} + 1$. □

3.2 Consequences and parity structure

Corollary 4 (Reduction). *Computing ρ_n reduces to computing σ_{n-1} .*

Remark 5 (Why $W^{(m)}$ and not $V_{(m-1,1)}$?). Naively one might hope $\rho_n = 1 + \rho_{n-1}$, which would yield $\rho_n = n - 2$. But this is wrong (it overshoots the true rank $\lceil (n-2)/2 \rceil$). The correct recurrence (3) restricts the inductive subspace to $W^{(n-1)}$ (omitting $v_{T(n-1)}$) because the bridge vector $\alpha_{n-1}v_{n-1} + \beta_{n-1}v_n$ in Y_n already supplies the v_{n-1} direction — so we do not need $W^{(n-1)} = V_{(n-2,1)}$. The codim-1 constraint on $W^{(m)}$ is the key reason the rank only grows by 1 every other step, not every step.

3.3 Verification at $n \leq 7$

The table below confirms (3) computationally (`verify_recurrence.py`, $q = 7 \in \mathbb{Q}$ in the seminormal basis):

n	ρ_n	σ_{n-1}	$1 + \sigma_{n-1}$	match?
3	1	0	1	✓
4	1	0	1	✓
5	2	1	2	✓
6	2	1	2	✓
7	3	2	3	✓

4 The conjectural rank formula

Conjecture 6 (Rank formula for $V_{(n-1,1)}$). *For $n \geq 2$,*

$$\rho_n = \left\lceil \frac{n-2}{2} \right\rceil = \left\lfloor \frac{n-1}{2} \right\rfloor. \tag{4}$$

Equivalently, $\sigma_m = \left\lceil \frac{m-3}{2} \right\rceil$ for $m \geq 2$.

Proposition 7 (Verified at $n \leq 7$). *Conjecture 6 holds for $n \in \{2, 3, 4, 5, 6, 7\}$ (see Section 3 verification, plus the trivial $\rho_2 = 0$ for $V_{(1,1)} = \text{sign rep at } S_2$).*

4.1 Reduction to a kernel-containment statement

A second equivalent form of the recurrence will be useful. Restating the proof of Theorem 3: the kernel of Π^{S_n} on $V_{(n-1,1)}$ is determined by

$$\ker \Pi^{S_n}|_{V_{(n-1,1)}} = (R'_n)^{-1}(\ker \Pi^{S_{n-1}}|_{V_{(n-1,1)}}),$$

and a direct computation in the seminormal basis (using Y_n from Theorem 1) shows

$$\rho_n = (n-2) - \dim(K_{n-1} \cap W^{(n-1)}), \quad (5)$$

where $K_{n-1} := \ker \Pi^{S_{n-1}}|_{V_{(n-2,1)}}$ at S_{n-1} .

Equivalently, $\sigma_{n-1} = (n-3) - \dim(K_{n-1} \cap W^{(n-1)})$, so (5) is the same as $\rho_n = 1 + \sigma_{n-1}$.

4.2 The dimension-of-intersection conjecture

If the rank formula $\rho_n = \lceil (n-2)/2 \rceil$ holds, then via (5) we read off the corresponding intersection dimension:

Conjecture 8 (Kernel intersects W with parity). *For $n \geq 3$, the kernel $K_{n-1} \subseteq V_{(n-2,1)}$ satisfies*

$$\dim(K_{n-1} \cap W^{(n-1)}) = \lfloor \frac{n-2}{2} \rfloor.$$

Equivalently,

- For n even: $K_{n-1} \subseteq W^{(n-1)}$ (the kernel has zero $v_{T(n-1)}$ -component).
- For n odd: $K_{n-1} \not\subseteq W^{(n-1)}$, and the intersection $K_{n-1} \cap W^{(n-1)}$ has codimension exactly 1 in K_{n-1} .

Verification at $n \leq 7$. By direct seminormal-basis computation:

n	$\dim K_{n-1}$	$\dim W^{(n-1)}$	$\dim(K_{n-1} \cap W^{(n-1)})$	predicted $\lfloor (n-2)/2 \rfloor$
3	1	0	0	0
4	1	1	1	1
5	2	2	1	1
6	2	3	2	2
7	3	4	2	2

For $n = 4$ ($K_3 = \text{span}(v_2)$, $W^{(3)} = \text{span}(v_2)$): $K_3 \subseteq W^{(3)}$. ✓ For $n = 5$ ($K_4 = \text{span}(v_2, c_1 v_3 + c_2 v_4)$ with $c_2 \neq 0$): $K_4 \not\subseteq W^{(4)} = \text{span}(v_2, v_3)$ since the second generator has a nonzero v_4 -component. ✓

Remark 9 (The structural meaning of the parity). The parity dichotomy in Conjecture 8 reflects whether $\Pi^{S_{n-1}}(v_{T(n-1)})$ is in the image $\Pi^{S_{n-1}}(W^{(n-1)})$. For n even (kernel contained), the $v_{T(n-1)}$ -direction projects *nontrivially* through Π , so it adds a new image direction. For n odd, the $v_{T(n-1)}$ -direction has a kernel component, so adding $v_{T(n-1)}$ to $W^{(n-1)}$ does *not* grow the image. This is the structural “every-other-step” rank growth pattern.

Remark 10 (Why the parity conjecture is hard). At first glance one might hope to prove Conjecture 8 by induction on n , identifying the kernel K_{n-1} via its own staircase structure. The difficulty is that K_{n-1} is the kernel of $\Pi^{S_{n-1}}$ (not of R'_{n-1}), so an inductive description requires understanding the “stable kernel” of the iterated staircase — a more delicate object than the image computed in Theorem 1. We do not yet have a clean closed form for K_{n-1} in the seminormal basis at arbitrary n .

5 Honest assessment

What is proved.

1. Theorem 1: a fully structural description of $R'_n(V_{(n-1,1)})$ as the explicit $(n-2)$ -dimensional subspace $\text{span}(v_2, \dots, v_{n-2}, \alpha_{n-1}v_{n-1} + \beta_{n-1}v_n)$. The proof is by induction on the partial product P_j and uses only the Hoefsmit-basis eigenvalue structure of each T_j on $V_{(n-1,1)}$.
2. Theorem 3: the rank recurrence $\rho_n = 1 + \sigma_{n-1}$, derived from Theorem 1 and the multiplicity-free branching $V_{(n-1,1)} \downarrow S_{n-1} = V_{(n-2,1)} \oplus V_{(n-1)}$.
3. The conjectural rank formula $\rho_n = \lceil (n-2)/2 \rceil$ is verified computationally for $n \in \{2, \dots, 7\}$ in the seminormal basis.

What is conjectural. The rank formula $\rho_n = \lceil (n-2)/2 \rceil$ for general n . By Theorem 3 and (5), the rank formula is equivalent to Conjecture 8: a parity-dependent statement on whether the kernel K_{n-1} of $\Pi^{S_{n-1}}|_{V_{(n-2,1)}}$ is contained in (or has codim-1 intersection with) the codim-1 subspace $W^{(n-1)}$ that omits the seminormal vector $v_{T(n-1)}$. The parity rule is verified at $n \leq 7$ but not yet proved structurally.

Why this is useful. Theorem 1 gives the first *fully structural* identification of $R'_n(V_{(n-1,1)})$ as an explicit subspace with explicit Hoefsmit-derived basis. Combined with the multiplicity-free branching, it converts the rank computation into a clean recurrence on σ_m values — reducing the rank-formula problem to an explicit codim-1 statement (Conjecture 8) about the kernel of $\Pi^{S_{n-1}}|_{V_{(n-2,1)}}$ relative to the specific direction $v_{T(n-1)}$. This is the first concrete leverage we have on ρ_n beyond direct computation.

Computational note. All matrix computations are in the seminormal basis using SymPy with $q = 7 \in \mathbb{Q}$ (generic for V_λ). Scripts are at

~/projects/scratch/2026-05-08-evening-Rn-on-hook/verify_Rn_image.py,
~/projects/scratch/2026-05-08-evening-Rn-on-hook/verify_recurrence.py.